



SHIVNETRA 47

SHIV-NETRA47 AIR DEFENCE SYSTEM PRIVATE LIMITED

AUTONOMOUS AIR SYSTEMS • SWARM INTELLIGENCE • TACTICAL AIR DEFENCE

MADE IN INDIA • ATMANIRBHAR BHARAT

Engineering Collaboration: BCN In-House Technology Team

PROGRAM BLUEPRINT & TABLE OF CONTENTS

SWARM ARCHITECTURE
& TECHNICAL DEFENCE INDEX

A comprehensive operational reference detailing sovereign autonomous swarm intelligence, heterogeneous air reconnaissance, tactical C2, and frontline border air defence architectures engineered to Indian Armed Forces requirements.

● 12 TECHNICAL SECTIONS

● 3 DEFENCE CAPABILITY PILLARS

▲ 6-PAGE SOVEREIGN DEFENCE BROCHURE

★ MAKE IN INDIA & IDEX COMPLIANT

PART I • SWARM C2

SWARM & OVERWATCH ISTAR

AUTONOMOUS SWARM INTELLIGENCE & CONTESTED MESH C2

01

EXECUTIVE VISION & DOCTRINE

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Forward Border Operations • Multi-Tier Defence

High-altitude border operational requirements and multi-tier sovereign air defence integration doctrine.

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HETEROGENEOUS SWARM P2P MESH

PAGE 04

Peer-to-Peer Mesh • Autonomous Patrol

Distributed peer-to-peer RF mesh routing, decentralized patrol logic, dynamic collision avoidance, and node discovery.

03

6-NODE VS 24-NODE DEPLOYMENT

PAGE 04

Tactical Patrol • Contested Saturation Screen

Scalable squad-level tactical patrol (6 units) up to wide-area contested border saturation screens (24+ units).

04

AUTONOMOUS BOUNDING OVERWATCH

PAGE 04

Cooperative Search • Dynamic Target Handoff

Decentralized bounding overwatch search, cooperative sector handoff, and attrition-tolerant node re-routing.

PART II • AIR PLATFORMS

NANO-SCOUT & OVERWATCH

SUB-500G MICRO SCOUTS TEAMING WITH 20KG RELAYS

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NANO-SCOUT MICRO-UAV (350G)

PAGE 05

Pocket Deployable • Ultra-Low Acoustic

Disposable micro air vehicle for ultra-low acoustic forward scout missions, squad reconnaissance, and perimeter breach detection.

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OVERWATCH TACTICAL RELAY (~20KG)

PAGE 05

Long-Endurance VTOL • High-Altitude Relay

High-altitude VTOL mothership relay maintaining airborne mesh connectivity and broad-area surveillance at 8,000+ ft AMSL.

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MULTI-SPECTRAL EO/IR PAYLOADS

PAGE 05

Dual HD Optical Zoom • LWIR Thermal Core

Stabilized daylight zoom cameras paired with uncooled microbolometer thermal imagers for 24/7 all-weather target tracking.

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SUB-ZERO HIMALAYAN ENVELOPE

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Cold-Soak Conditioning • Mountain Wind Handling

Active battery thermal management, anti-icing carbon composite rotors, and sustained operation in high mountain wind gradients.

PART III • BATTLESPACE C2

TACTICAL GCS & HARDENING

HARDENED BATTLESPACE C2, NAVIC TELEMETRY & PRODUCTION

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UNIFIED TACTICAL GCS WORKSTATION

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Ruggedized Console • Multi-UAV Sortie

Ruggedized military operator console managing simultaneous multi-UAV swarm sorties, live video feeds, and mission profiles.

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EW HARDENING & NAVIC TELEMETRY

PAGE 06

Anti-Jam Frequency Hopping • NAVIC Sync

Anti-jamming frequency-hopping RF data-links, sovereign NAVIC satellite synchronization, and optical dead-reckoning fallback under EW denial.

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SORA SAIL II/III SAFETY STANDARDS

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DGCA / Military Clearances • FTS Mitigation

Specific Operations Risk Assessment (SORA SAIL II/III), emergency flight termination (FTS), and sovereign military airspace clearances.

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INDIGENOUS PRODUCTION & CONTACTS

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Composite Autoclave • IDEX / MoD Procurement

Sovereign carbon-composite fabrication, localized component assembly, and IDEX / MoD procurement roadmap for scalable Indian production.

SOVEREIGN DOCTRINE

UNIFIED AIR ARCHITECTURE

One tactical ecosystem linking micro swarm scouts, overwatch relays, and ground control under synchronized command.

• STANAG 4586 Interoperable

SWARM AUTONOMY

DECENTRALIZED MESH LOGIC

Autonomous collaborative search patterns, dynamic node re-routing, and seamless target handoff without single point of failure.

• Distributed AI Engine

CONTESTED SPECTRUM

NAVIC • MESH ENCRYPTION

Resilient anti-jam frequency hopping with autonomous optical dead-reckoning fallback under intense GNSS spoofing or jamming.

• Dual-Band NAVIC • INS

MAKE IN INDIA

100% SOVEREIGN IPR & MRO

Domestic carbon-composite production and localized depot-level lifecycle maintenance aligned with Defence Acquisition Procedure (DAP).

• IDEX / AITIT Compliant

MULTI-DOMAIN UNMANNED AERIAL ECOSYSTEM • DEFENCE SCALING ROADMAP

Phased Deployment: Prototyping & High-Altitude Trials → MoD User Evaluation → Sovereign Induction

● 12 TECHNICAL SECTIONS

● 3 OPERATIONAL PILLARS

★ TRL 7+ TESTED

■ SOVEREIGN IPR